Minjae V. Kim

Department of Physics and Astronomy, University of Pennsylvania
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Education

University of Pennsylvania — *Philadelphia, PA*
Ph.D. Physics. — *Aug 2026 - Present*

- Advisor: Prof. Jonathan J. Heckman

Johns Hopkins University — *Baltimore, MD*
B.Sc. Physics and Mathematics. Minor: Visual Arts. — *Sep 2020 - May 2025*

Sejong Science High School — *Seoul, Republic of Korea*
Feb 2017 - Feb 2020

Publications/Preprints

M. V. Kim, J. H. Krolik, and S. C. Noble. "Effects of Physical Cooling on the Structure of Circumbinary Disks." under review in ApJ *arXiv:2609.21136 [astro-ph.HE]*

T. Waddleton, **M. V. Kim**, and I. Bah. "Liquid Crystals and Generalized Symmetries." *in preparation.*

M. Chitoto, P. Jefferson, and **M. V. Kim**. "Towards an EFT of the black D0-brane near horizon." *in preparation.*

Presentations

"Symmetries of Wick Rotated QFT", *Field Theory and Topology Reading Seminar, 2026, UPenn*

"Magnetohydrodynamics of an accretion disk with radiation", *Physics Undergraduate Research Talks, Sep 2023, JHU*

"Simulation of a black hole accretion disk", *Center for Astrophysics Research Experience Summer 2022, Aug 2022, JHU*

Awards

NSF GRFP Honorable Mention, NSF — *2025*
SOAR Summer Plunge Scholarship, JHU — *2024*
William Lowell Putnam Award, JHU Department of Mathematics — *2024*
Society of Physics Students Distinguished Chapter Award — *2022 - 2023*

Teaching

Department of Physics and Astronomy, University of Pennsylvania

Lab Instructor, General Physics: Mechanics (PHYS 0150) — *Fall 2026*

Department of Physics and Astronomy, JHU

Learning Assistant, General Physics II — *Spring 2024*
Learning Assistant, Classical Mechanics I — *Fall 2023*
Learning Assistant, General Physics I — *Fall 2022*
Learning Assistant, General Physics II — *Spring 2022*

Learning Assistant, General Physics I *Fall 2021*

Outreach and Service

Volunteer, Science After Hours, The Franklin Institute *2026*
Department of Physics and Astronomy, University of Pennsylvania

Volunteer, Annual Physics Fair *2025*
Department of Physics and Astronomy, JHU

Volunteer, Peer Mentoring and Academic Support *2021 - 2025*
Department of Physics and Astronomy, JHU

Volunteer, Telescope Maintenance for Public Observing *2022 - 2023*
Maryland Space Grant Observatory, JHU

Technical Skills

Languages: Korean (Native), English (Fluent)

Software: C, C++, Mathematica, Python, LaTeX, MPI, Shell Scripting, GNU/Linux, Windows